

Comparison with human and machine generating language and its applications

2025. 06. 30

Hyungbin Park, Kihoon Lee, Gwangsu Kim



전북대학교
JEONBUK NATIONAL UNIVERSITY



JBNU G·LAMP
Project Group



물질에너지초과학연구소
Research Institute for Materials and Energy Sciences

**A Comparative Analysis of Large Language
Model(LLM) and Human-Generated Text
Using Natural Language Processing**

Contents

1. Introduction
2. Data & Methodology
3. Syntactic Fingerprinting
4. Topical & Lexical Signatures
5. Classification
6. How to use LLM about RIMES ?
7. Conclusion

1. Introduction

Can we still tell apart human and machine-generated text?

The line between Human and AI is fading fast

- Modern Large Language Models like GPT and Gemini now produce content **nearly indistinguishable** from human writing.
- These models can convincingly mimic human **style, tone, and reasoning** in a wide range of contexts.
- This remarkable advancement raises a **critical challenge**: ensuring the **reliability and authenticity** of digital content.
- Without clear classification, misinformation, AI-generated spam, and academic dishonesty may spread unchecked.
- Such risks affect vital areas like education, journalism, and cybersecurity, where trust is essential.
- Therefore, **distinguishing between human and machine-generated text** is no longer optional, but necessary

Classifying text origin is essential to ensure trust, safety, and ethical AI use

1. Introduction

Related work about text classification Based Natural Language Processing

D. Mamata, K. Selvakumar , P.J.A. Alphonse, A **Comparative** Study on **TF-IDF** feature Weighting Method and its Analysis using Unstructured Dataset, arXiv:2308.04037 (2023)

Kim, SW., Gil, JM. Research paper **classification systems** based on **TF-IDF** and **LDA** schemes. Hum. Cent. Comput. Inf. Sci. 9, 30 (2019)

Essa, E., Omar, K. & Alqahtani, A. **Fake news detection** based on a hybrid BERT and **LightGBM** models. Complex Intell. Syst. 9, 6581–6592 (2023).

M. H. Ibnath, M. Sheikh, M. Sheikh and J. F. Shormy, **Enhanced Suicidal Ideation Detection** via **Text Classification** and Ensemble Methods, 2025 International Conference on Electrical, pp. 1-6 (2025).

Voskergian, D., Jayousi, R. & Yousef, M. Topic selection for **text classification** using ensemble **topic modeling** with grouping, scoring, and modeling approach. Sci Rep 14, 23516 (2024).

K. H. Lee, **Natural Language Processing Based Comparative Analysis of Research, News, and Complaints Data** in the Cadastre Domain, ksc vol.41, no.1, pp.83-97 (2025).

Essa, E., Omar, K. & Alqahtani, A. **Fake news detection** based on a **hybrid BERT** and **LightGBM** models. Complex Intell. Syst. 9, 6581–6592 (2023).

2. Data & Methodology

TF-IDF

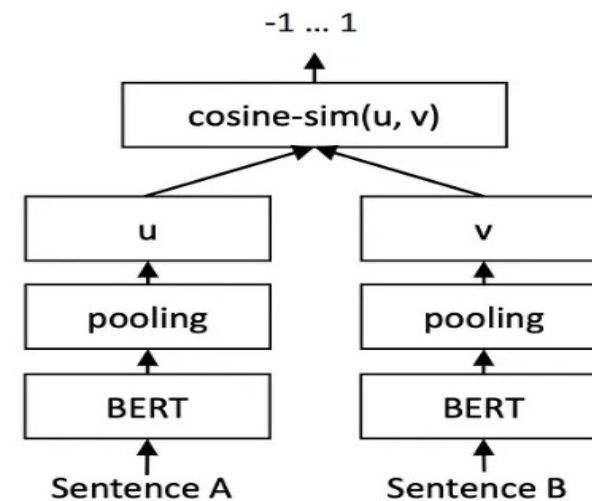
$$TF(t, d) = \frac{(\text{Number of occurrences of term } t \text{ in document } d)}{(\text{Total number of terms in the document } d)}$$

$$IDF(t, D) = \log_e \frac{(\text{Total number of documents in the corpus})}{(\text{Number of documents with term } t \text{ in them})}$$

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

Term frequency (TF) multiplied by inverse document frequency (IDF) to represent how important a word is in a particular document

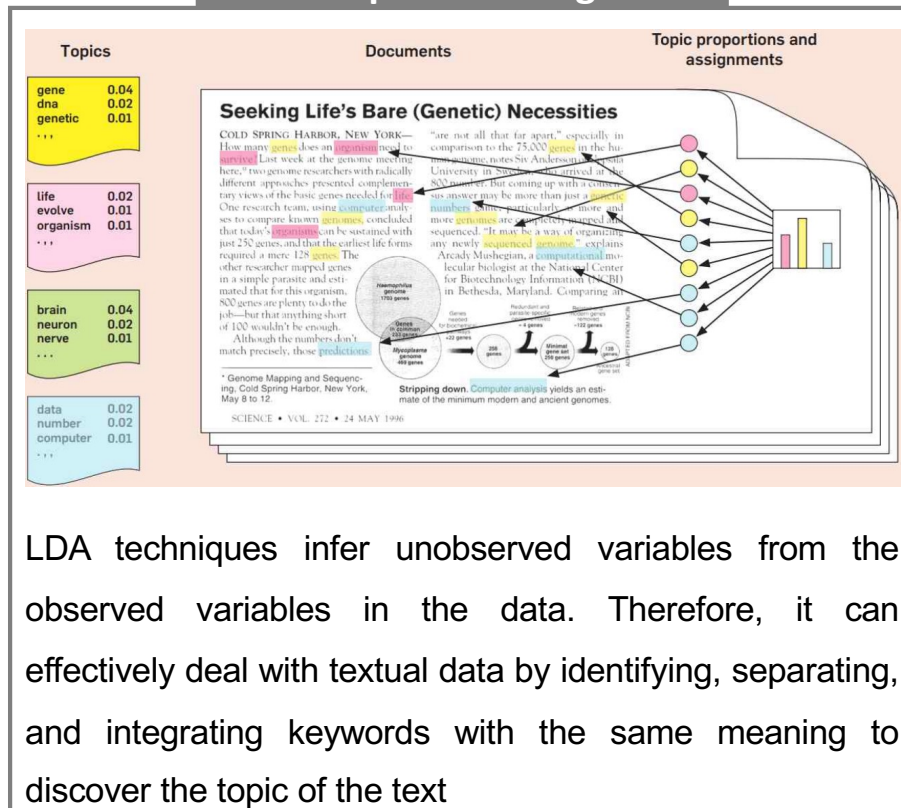
Sentence Bert & Cosine Similarity



SBERT architecture at inference, for example, to compute similarity scores. This architecture is also used with the regression objective function

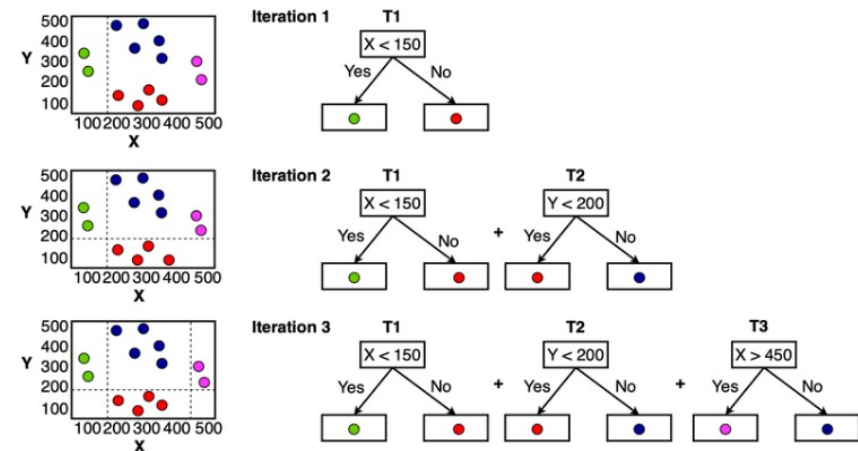
2. Data & Methodology

Topic Modeling



LDA techniques infer unobserved variables from the observed variables in the data. Therefore, it can effectively deal with textual data by identifying, separating, and integrating keywords with the same meaning to discover the topic of the text

LightGBM



Algorithms use an iterative process called boosting, which sequentially improves the base learner to build a resilient predictive model. By leveraging gradient descent to optimize the loss function during each iteration, increasingly accurate models can be developed

2. Data & Methodology

Data Collection : Building a Diverse Corpus

Human Text

Quora

A Place to share Knowledge and
Better understand the world



~2800 responses collected from Quora
(up to 30 answers per question)

LLM Text

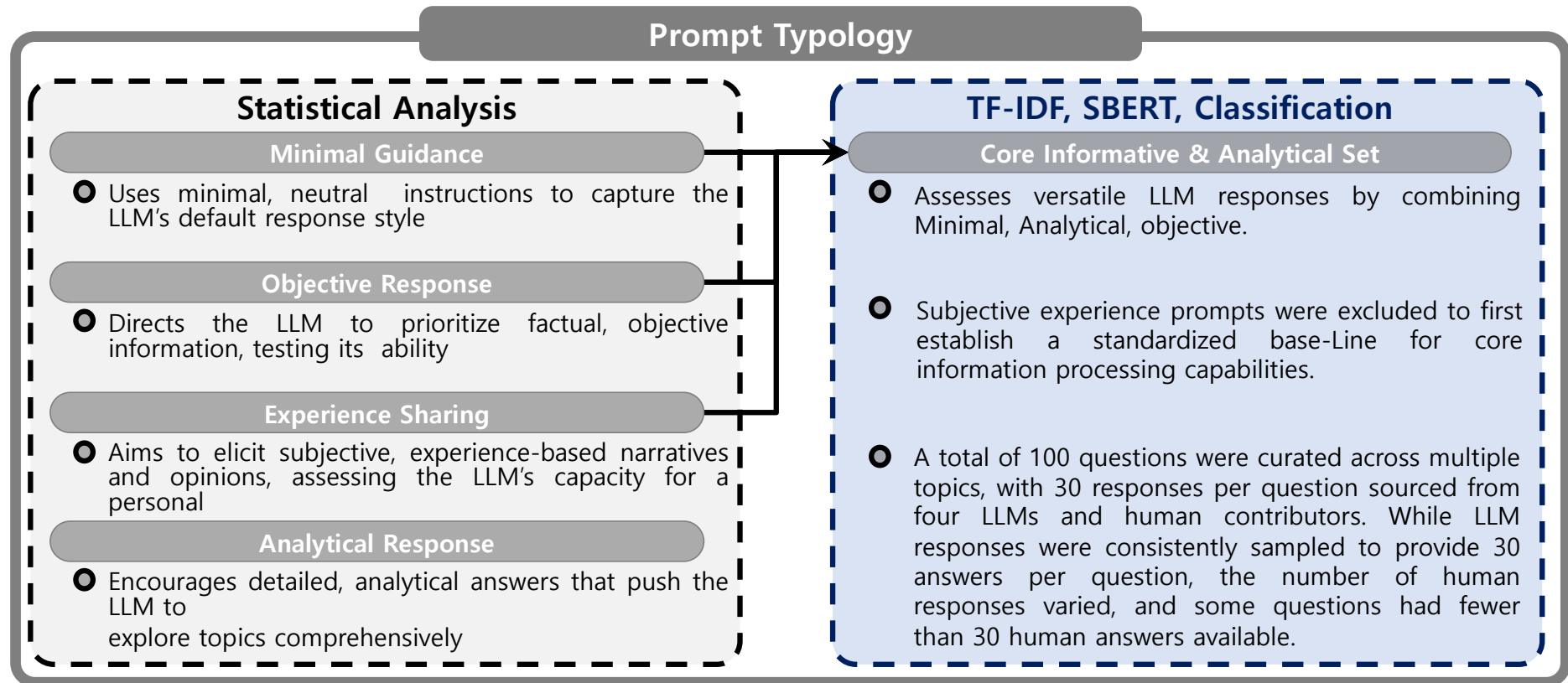


3000 responses from each of the four LLMs
(30 answers x 100 questions)

Curated a set of 100 unique questions spanning a diverse range of topics to ensure the generality

2. Data & Methodology

Data Collection : Building a Diverse Corpus



2. Data & Methodology

Use NLP techniques to distinguish between human and LLM generated text

Step.1 Data Collection and Processing

- Curated a dataset consisting of 100 questions and 30 diverse answers, standardized by NLTK preprocessing for uniform token count

Step.2 Part of Speech(POS)

- stylometry and Part-of-Speech (POS) analysis are utilized to measure the fundamental structure and grammatical characteristics of the text

Step.3 TF-IDF

- lexical signature analysis via TF-IDF identifies unique and significant keywords associated with each author

Step.4 Similarity analysis - Sentence-Bert

- semantic embeddings employing Sentence-BERT (SBERT) capture contextual meanings and semantic similarities within texts

Step.5 Hierarchical LDA-Based Subtopic Modeling

- topic modeling using Latent Dirichlet Allocation (LDA) explores hidden topics and underlying themes present in the texts

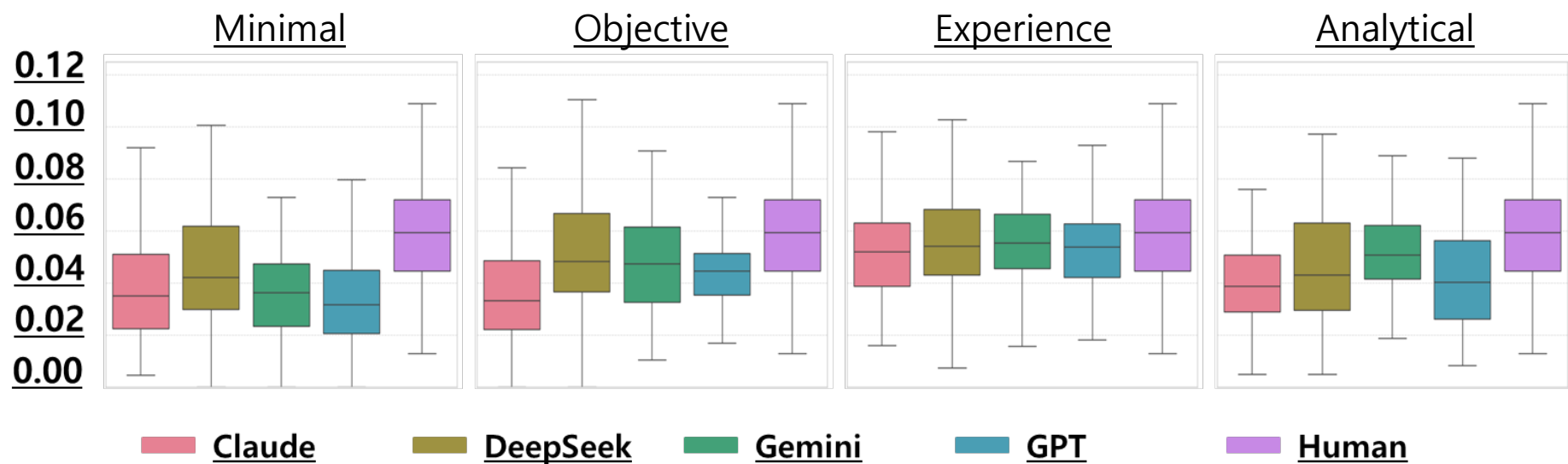
Step.6 Classification - LightGBM

- these derived features are integrated into a classification model trained with LightGBM to predict whether a text was authored by a human or a specific LLM

3. Syntactic Fingerprinting

How They Speak (1) – Who writes “me” a lot, and what will they talk about?

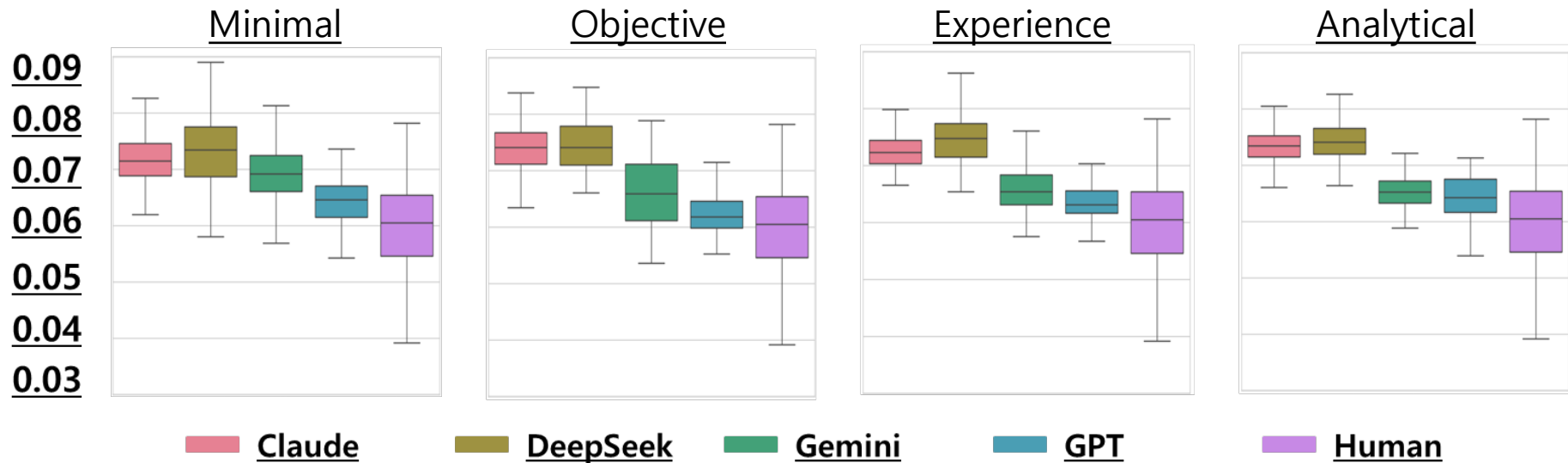
1. The average usage proportion is statistically higher in human responses
2. This suggests humans directly express their personal experiences, opinions, and emotions



3. Syntactic Fingerprinting

How They Speak (2) - How Diverse is Their Word Choice?

1. DeepSeek (avg. 0.75) and Claude (avg. 0.73) exhibit relatively high lexical density, indicating efficient use of diverse words relative to their answer length
2. In contrast, GPT (avg. 0.64) and Human (avg. 0.61) show a somewhat lower lexical density



4. Topical & Lexical Signatures

TF-IDF – Which word they choose ?

1. TF-IDF identifies uniquely important keywords by measuring how often a word appears in a document versus how rare it is across all documents
2. This lexical gap indicates a fundamental difference : LLMs function as generalist information processors, while human act as specialists of their own experience

$$TF - IDF = TF * IDF$$

Top-10	Human		GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score	Word	Score
1	One	0.2570	Like	0.3072	Like	0.3796	Like	0.3088	Like	0.2530
2	Time	0.2424	One	0.2373	Think	0.2711	One	0.2925	People	0.2247
3	Would	0.2235	Time	0.1840	Thing	0.2285	People	0.2696	I've	0.2189
4	People	0.1879	People	0.1837	One	0.1966	Even	0.2426	Think	0.2042
5	Like	0.1807	Think	0.1768	People	0.1805	Way	0.1815	Something	0.1804
6	Get	0.1582	Might	0.1735	Even	0.1754	Thing	0.1621	One	0.1750
7	Day	0.1550	Thing	0.1595	Okay	0.1751	Time	0.1506	Actually	0.1681
8	Year	0.1521	I've	0.1563	Really	0.1610	Make	0.1402	Thing	0.1529
9	Life	0.1457	Even	0.1517	Know	0.1607	Something	0.1389	Time	0.1517
10	dont	0.1381	Life	0.1473	something	0.1361	dont	0.1331	someone	0.1344

4. Topical & Lexical Signatures

Sentence BERT – How similar is their context ?

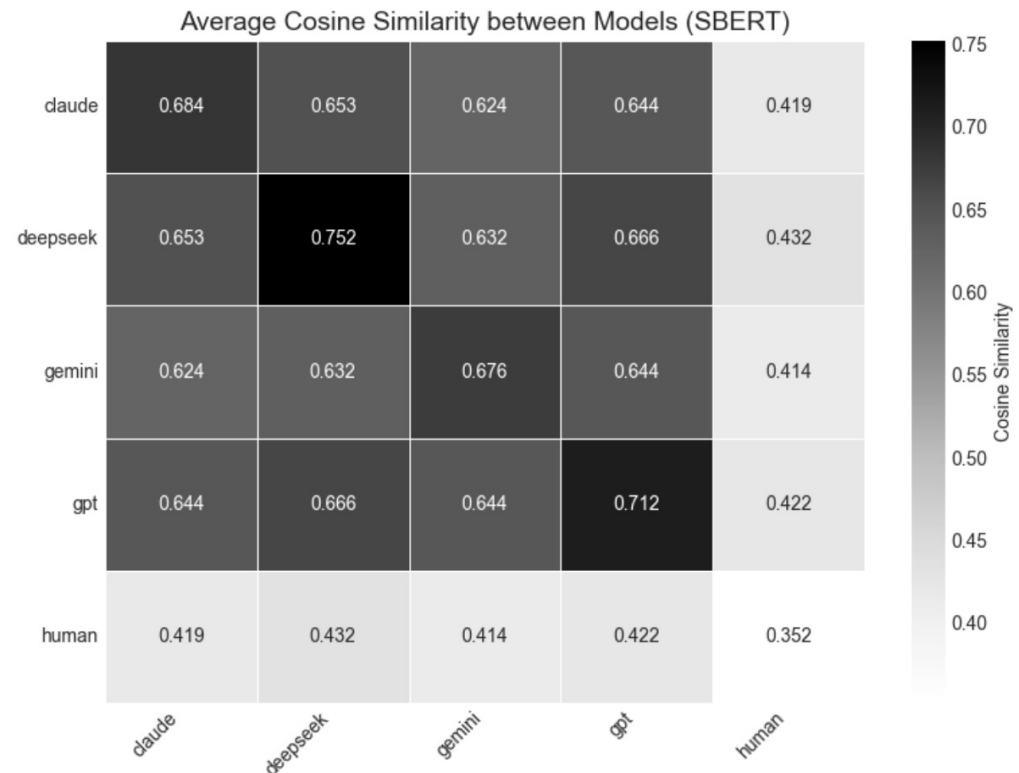
1. LLMs show high semantic similarity :

They consistently provide answers with similar meanings, reflecting a convergent thought process

2. Humans show high semantic

diversity : Their responses are distinct from LLMs and highly varied among themselves, reflecting unique individual perspectives

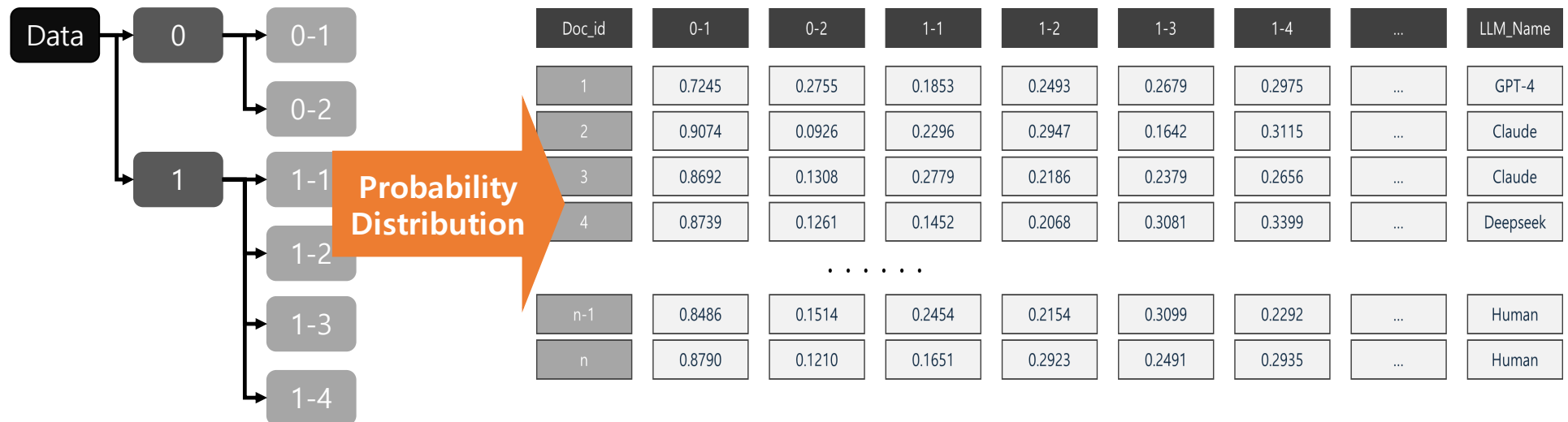
$$\text{similarity} = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i \times B_i}{\sqrt{\sum_{i=1}^n (A_i)^2} \times \sqrt{\sum_{i=1}^n (B_i)^2}}$$



5. Classification

Hierarchical LDA – Based Subtopic Modeling

1. To capture nuanced thematic structure, each text was transformed into a vector of subtopic probabilities
2. This vector served as the sole input, allowing us to predict authorship based on topic composition

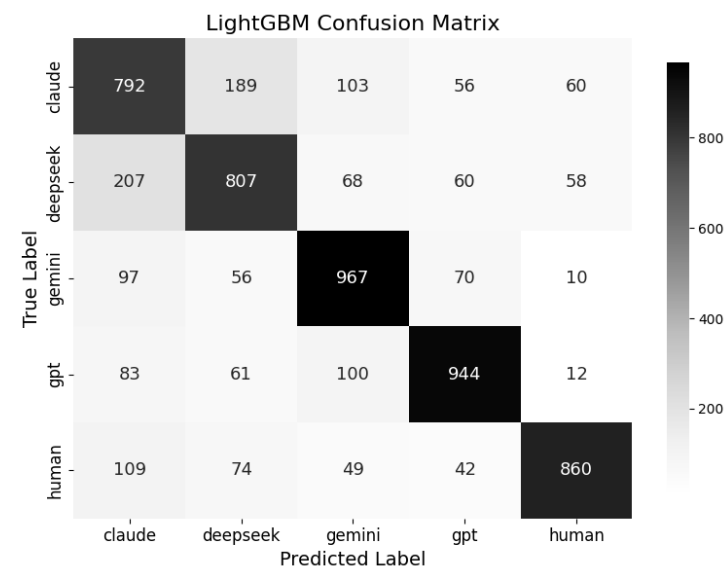


5. Classification

LightGBM Based Classification

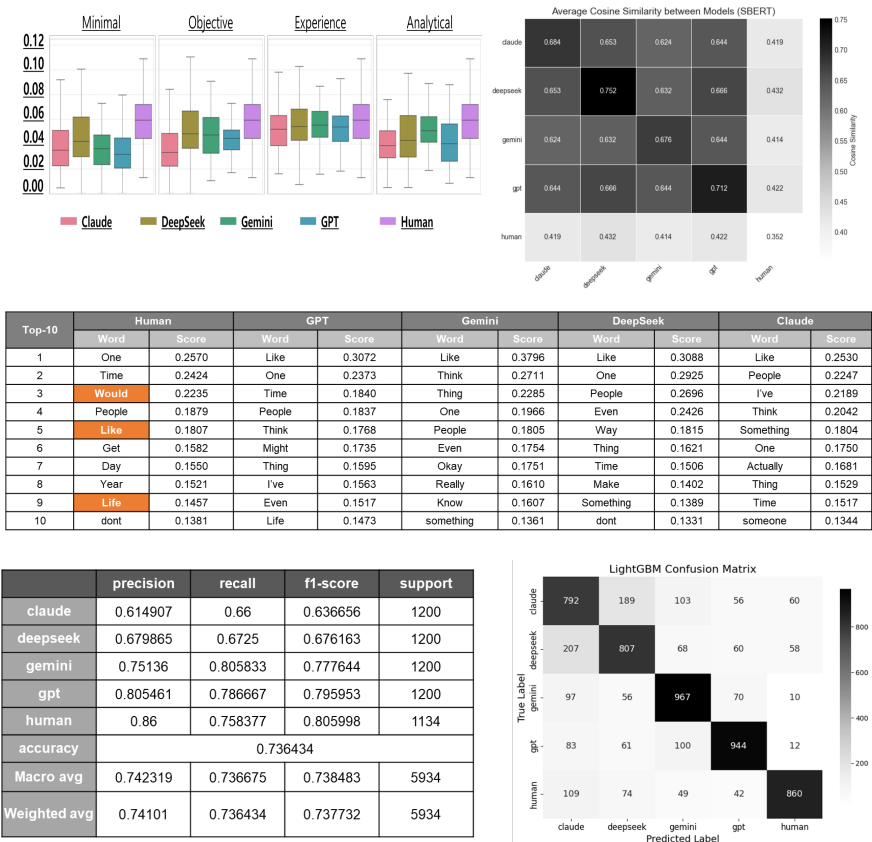
1. Our model achieved 73.6% accuracy using only subtopic features, proving thematic structure is key
2. It classified human text most accurately (F1-score : 0.81) and revealed distinct LLM patterns

	precision	recall	f1-score	support
claude	0.614907	0.66	0.636656	1200
deepseek	0.679865	0.6725	0.676163	1200
gemini	0.75136	0.805833	0.777644	1200
gpt	0.805461	0.786667	0.795953	1200
human	0.86	0.758377	0.805998	1134
accuracy	0.736434			
Macro avg	0.742319	0.736675	0.738483	5934
Weighted avg	0.74101	0.736434	0.737732	5934



Conclusion

- First, our research confirms that measurable linguistic patterns clearly differentiate human and LLM generated text
- Second, these distinctions arise from fundamental differences in semantic diversity, narrative style, and syntax
- Lastly, this work provides a crucial tool to combat “Model Collapse” and promote a sustainable AI ecosystem



6. How to use LLM about RIMES ?

Related work about Materials Science using LLM

Kumar, Kumar, **Generative AI in Materials Science: Accelerating Discovery Through Inverse Design** (May 12, 2025). Available at SSRN:.

Henry W. et al, Monte Carlo Thought Search: **Large Language Model** Querying for Complex Scientific Reasoning in **Catalyst Design**, The 2023 Conference on Empirical Methods in Natural Language Processing

Zi-Yi Chen et al, MatChat: A **large language model** and application **service platform for materials science** 2023 Chinese Phys. B 32 118104

Z. Pei, J. Yin, J. Zhang, **Language models** for materials discovery and sustainability: Progress, challenges, and opportunities, Progress in **Materials Science**, Volume 154, 2025

6. How to use LLM about RIMES ?

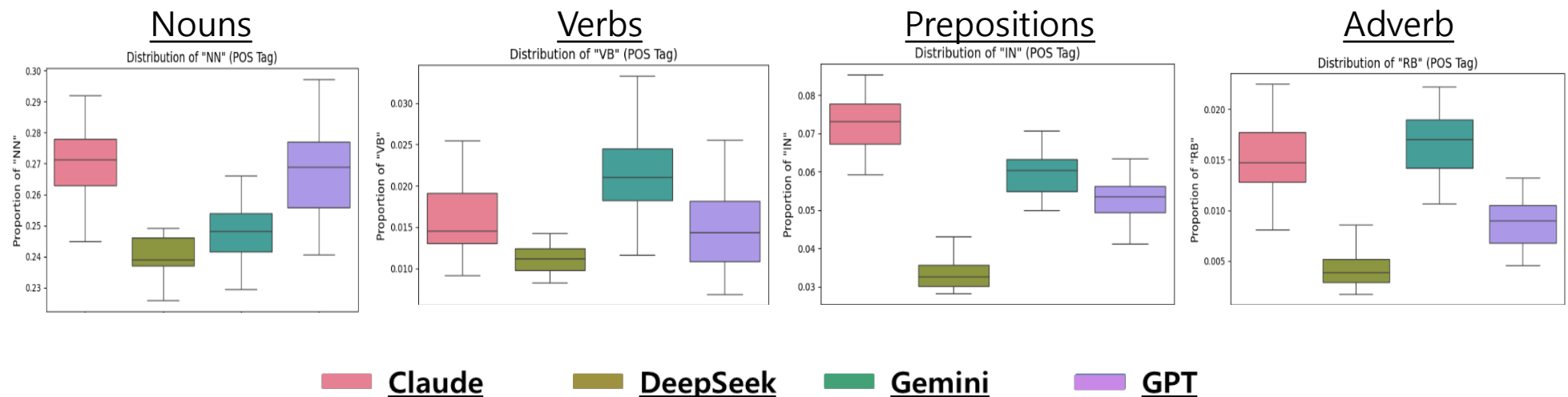
Data

- Curated a dataset consisting of 1 question and 30 diverse answers from each model
- Question – I require a detailed overview of synthesis methods of five key energy storage materials
1. LLZO 2. LFP 3. $VOSO_4$ 4. Na-Sodium-Sulfur 5. MoS_2
provide : A) A description of two common synthesis methods.
B) A brief explanation of the core principle or advantage of each synthesis method
- Answer(GPT)
1. $Li_7La_3Zr_2O_{12}$ (LLZO)
Solid-State Reaction Method
 - Process: This involves reacting precursor powders of Li_2CO_3 , La_2O_3 , ZrO_2 , and sometimes Al_2O_3 (for doping) by thorough mixing, followed by high-temperature calcination.
 - Advantages: Well-established method suitable for large-scale production. It allows formation of a highly crystalline phase.

6. How to use LLM about RIMES ?

POS analysis

1. Claude and GPT tend to use more nouns (NN) for descriptive explanations, while Gemini prefers verbs
2. Claude's high use of prepositions (IN) suggests complex sentences
3. Gemini's adverb (RB) use adds rich detail



6. How to use LLM about RIMES ?

TF-IDF

1. GPT is descriptive and process-oriented, while Gemini emphasizes specific material names and properties
2. These patterns reflect their distinct strengths in technical explanation, practicality, materials, and theory

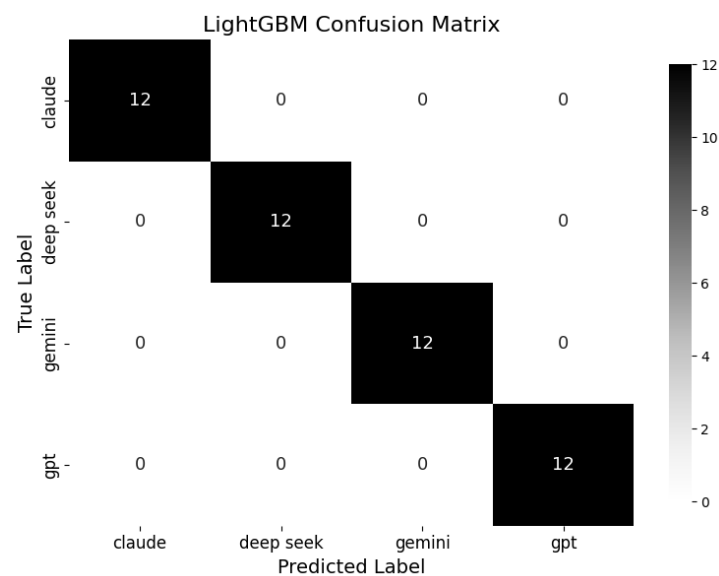
Top-10	GPT		Gemini		DeepSeek		Claude	
	Word	Score	Word	Score	Word	Score	Word	Score
1	Description	0.1369	Eg	0.1690	Advantage	0.2006	Core	0.1119
2	Advantage	0.0967	Sulfur	0.1188	Process	0.1427	Core Principle	0.0991
3	Core	0.0899	Carbon	0.1122	Hydrothermal	0.0827	Principle	0.0887
4	Principle	0.0847	Principle	0.0938	Hso	0.0773	Common synthesis	0.0884
5	Mos	0.0662	Description	0.0916	Mos	0.0725	Advantage	0.0871
6	Hydrothermal	0.0623	Mos	0.0760	Challenge	0.0667	Common	0.0686
7	Sulfur	0.0579	Core	0.0706	Eg	0.0662	Carbon	0.0668
8	Process	0.0545	Source	0.0705	Sulfur	0.0659	Involves	0.0583
9	ambient	0.0499	Hours	0.0668	Melt quenching	0.0655	Oxidation	0.0507
10	sodium	0.0464	Hydrothemat	0.0663	Scalable	0.0643	Reduction	0.0499

6. How to use LLM about RIMES ?

Classifying LLMs through a Classification Model

1. For a specific materials query, our model identified the source LLM with 100% accuracy
2. This proves each LLM has a distinct style, making them suited for different, purpose-driven tasks

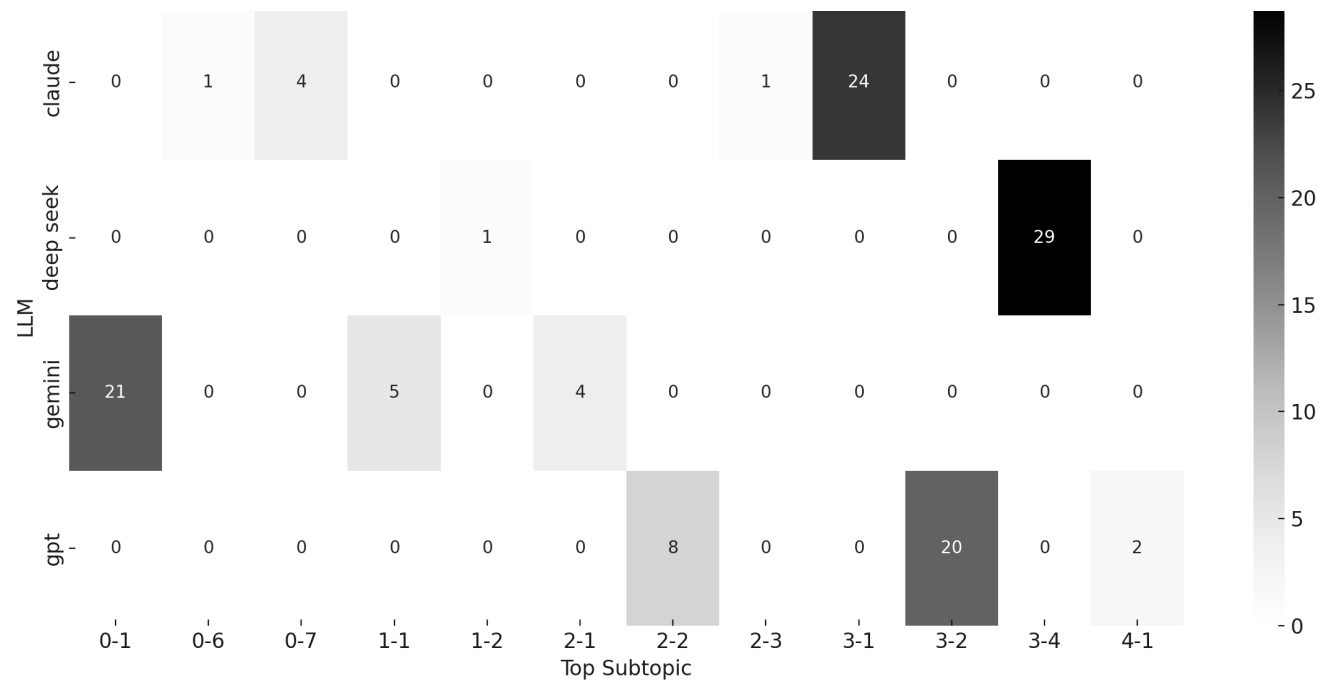
	precision	recall	f1-score	support
claude	1	1	1	12
deep seek	1	1	1	12
gemini	1	1	1	12
gpt	1	1	1	12
accuracy	1			
macro avg	1	1	1	48
weighted avg	1	1	1	48



6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(1)

1. This heatmap visualizes the distribution of the subtopics selected as the top topics in the responses generated by each LLM (GPT, Claude, Gemini, and DeepSeek)



6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(2) – Weight colculation

$$\sum_{d \in D_{\text{LLM}}} \sum_{k \in K_{t_d}} w_{k,t_d}$$

Cumulative weighted sum of keywords by LLM

D_{LLM} - The set of documents generated by that LLM
 t_d - Subtopics with the highest probability in Doc d
 K_{t_d} - List of keywords in the topic t_d
 w_{k,t_d} - Weight assigned to keyword K in topic t_d

Gemini

- Synthesis(0.647), sulfur(0.644), carbon(0.603), method(0.517), vanadium(0.442), principle(0.373), high(0.368), temperature(0.322)

ChatGPT

- Synthesis(1.012), advantage(0.645), high(0.624), method(0.586), core(0.518), state(0.494), solid(0.456), reaction(0.446)

Claude

- Method(0.706), synthesis(0.680), core(0.620), principle(0.601), methods(0.550), advantage(0.529), carbon(0.474), solid(0.422)

DeepSeek

- Synthesis(1.091), advantage(0.967), process(0.960), high(0.757), methods(0.727), solid(0.666), state(0.637), hydrothermal(0.513)

6. How to use LLM about RIMES ?

Interpreting results with subtopics and probability distribution matching(3)

Gemini (sulfur, carbon, vanadium, method)

- Tend to describe material properties and synthesis procedures in a balanced manner

ChatGPT (synthesis, advantage, method, high)

- Strong at comparing synthesis methods and analyzing their technical effects

Claude (principle, method, core, advantage)

- Values conceptual understanding, focusing on core principles and methods

DeepSeek (process, scalable, hydrothermal)

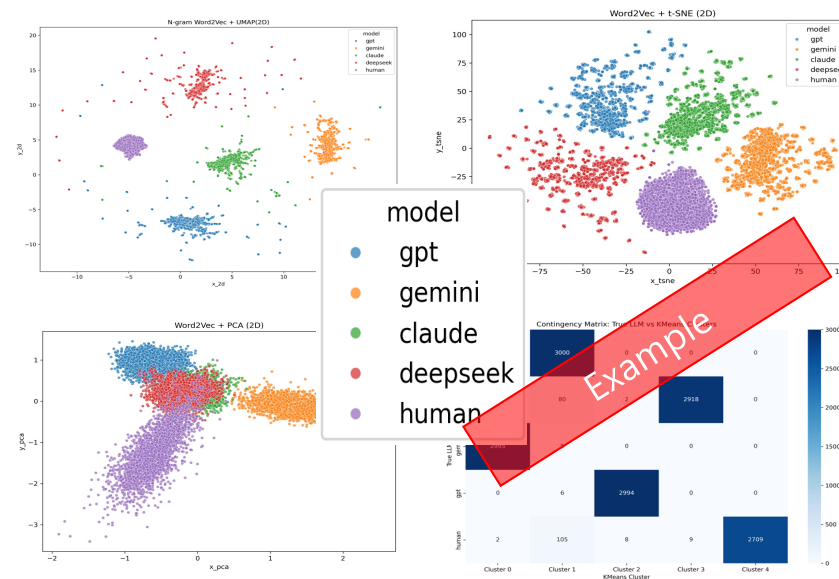
- Focuses on practical, process-oriented explanations of synthesis methods

7. Conclusion

Summary

- Our results show each LLM has consistent and distinct topical preferences for materials science tasks
- For example, GPT focuses on synthesis procedures, while Claude emphasizes conceptual principles
- This offers a measurable basis to strategically deploy the right LLM for specific research goals

Future Works



- We will improve the identification of LLM-specific characteristics and study reinforcement learning methods for **RIMES**



Thank you for your attention

Do you have any question??



전북대학교
JEONBUK NATIONAL UNIVERSITY



JBNU G·LAMP
Project Group



물질에너지초과학연구소
Research Institute for Materials and Energy Sciences